

DSS5105 Assignment-1

SG Private Housing Transaction Prediction

Shreya Sriram

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1 Analysis

1.1 Top 3 Key Factors

Based on SHAP (SHapley Additive exPlanations) analysis of the XGBRegressor model, the following three factors have the highest mean absolute contribution to transaction price:

- **Area (SQFT)** (SHAP: 0.1683) - Property size is the primary price driver, with larger units commanding proportionally higher values, reflecting Singapore's space premium.
- **Project Name** (SHAP: 0.1076) - Captures developer brand premium and development quality. Established projects command higher prices due to reputation and amenity standards.
- **Type of Sale** (SHAP: 0.1013) - Distinguishes primary market (new launches) from secondary market (resale), reflecting different pricing dynamics. New launches typically command premiums due to modern amenities and market positioning.

1.2 Patterns & Trends

- **Location premium:** Prime districts (6, 7, 9, 10 - CCR areas) command 2.5-3x prices compared to OCR districts (17, 18, 27, 28), demonstrating Singapore's location-based pricing tiers [Section 2.6].
- **Seasonal effects:** Q4 (especially November) shows 8-12% price premiums with peak transaction volumes, suggesting strategic timing for sellers [Section 3.2].
- **Tenure impact:** Long-term leasehold dominates (59.6%), followed by freehold (29.1%). Freehold properties command an 18.5% median price premium (\$1.80M vs \$1.52M) despite strong functional equivalence for buyers with 20-30 year horizons [Section 3.1].

- **Market segmentation:** 71.2% of properties fall under \$2M, with only 2.3% exceeding \$5M, indicating concentrated mid-market activity [Fig 4].

1.3 Buyer Recommendations

- **Target emerging districts:** Focus on districts 17, 18, 27, 28 offering average \$1.2M properties, representing 50%+ savings versus premium districts 6, 7, 9, 10 [Section 2.6].
- **Time purchases strategically:** Avoid November (peak prices). Purchase during August, December, or January when median prices drop 5-8% due to lower demand during festive seasons and school year transitions [Section 3.2].
- **Optimize size:** Target the 1,000-1,500 sqft range for optimal value at \$1,535 per sqft. Avoid units under 500 sqft (scarcity premium at \$1,746 per sqft) and properties over 5,000 sqft (luxury premium at \$2,281+ per sqft) unless specific needs justify the cost. [Section 2.5].